



Responses of linseed genotypes to salinity stress during early seedling growth

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Received: 10 October 2024, revised: 25 October 2024, accepted: 10 November 2024, DOI: <https://doi.org/10.59619/ej.6.2.3>

Abstract

Salinity is an important abiotic stress causing severe reduction in early seedling growth, plant stand and crop yield. Linseed (*Linum usitatissimum* L.) is known to be one of the oldest versatile crops having medium salt tolerant characteristics. The study aimed to evaluate the effects of salinity stress induced by NaCl on the early seedling growth of 29 linseed genotypes under 5 salinity levels (0, 4, 8, 12 and 16 dS m⁻¹) during the winter season of 2019-2020. The results showed that early seedling growth related parameters such as fresh and dry weights, and root length and shoot length of seedling, root to shoot ratio, seedling vigor index and salt tolerant index were decreased gradually with the increasing salinity level. The reduction in these traits was positively and significantly correlated with salinity levels. Compared to control, NaCl treatments of 4, 8, 12 and 16 dS m⁻¹ caused reduction in seedling dry weight, fresh weight, shoot length by (33.82, 46.08, 54.14 and 77.34%), (11.69, 25.35, 38.31 and 51.97%), (36.36, 50.00, 54.55 and 77.27%), and (15.30, 24.76, 39.69 and 50.77%), respectively. Root to shoot ratio was more or less similar in 7 genotypes, while it was significantly increased in 6 genotypes at all salinity levels. The ratio was increased significantly over control at 16 dS m⁻¹ in other 7 genotypes, and at 12 and 16 dS m⁻¹ in 9 genotypes. In 29 genotypes, the average reduction in seedling vigor index was 37.27, 52.21, 64.28 and 84.57% respectively at 4, 8, 12 and 16 dS m⁻¹. The highest mean seedling vigor index (SVI) was recorded from Neela followed by Lin-W-17, Lin-503, Lin-1403 and Lin-2403, while the lowest was found in BD-7143 (6.06) followed by Chilmari (6.46) and BD-7142. The average salt tolerant index of dry weight of whole seedling varied from 52.28 to 54.84% among 29 genotypes. Ten genotypes showed more than 60% tolerant level. Considering high level of early seedling growth and salt tolerance genotypes Neela, Lin-W-17, B-10708, BD-10710, Lin-1303, Lin-2403, BD-10698, Lin-S-19, BD-7143 and Lin-503 were noted as prospective materials for the development of salt tolerant linseed varieties.

Keywords: Linseed, flax, salinity tolerance, early seedling growth, genotype evaluation

Introduction

Linseed (*Linum usitatissimum*), a member of the family Linaceae is an economically potential versatile crop throughout the world (Berti *et al.* 2010). It is grown for oil seed as well as fiber or both. Its seeds contain 40-45% oil (EL-Nagdy *et al.* 2010). Linseed oil is used to manufacture paints, varnishes, clothes, patent leathers, printer inks, enamels, soaps, linen fabrics, linen threads, canvas bags, quality papers, etc. Linseed oilcake is a good manure and nutritious livestock feed. Its seeds are rich in protein, and essential fatty omega-3 fatty acids, including alpha-linolenic acid. Seed may be used for treatment of constipation, diabetes, high cholesterol, obesity, and swelling of the kidneys in people (Vittal *et al.* 2005).

Linseed can grow in a wide range of climatic conditions because it is comparatively less affected by stressful

environments. In spite of glycophytic crop plant, linseed is moderately tolerant to salinity (Day *et al.* 2018, Dubey *et al.* 2020). There is a prospect to grow the linseed in the Southern Coastal Zones of Bangladesh, where salinity is a problem (Ahsan 2010). To promote its cultivation, it is crucial to develop high-yielding linseed varieties with enhanced salt tolerance, which are currently lacking.

Salinity is a major abiotic stress which reduces crop yield through negative impacts on seed germination, seedling establishment, and plant growth (Munns and Tester 2008, Nasri *et al.* 2017). This stress alters different metabolic processes of plants, particularly photosynthesis and reduces the chance of survival. While some plants are sensitive to saline conditions, some survive by genetically controlled tolerance mechanisms induced by various

physiological, biochemical and molecular responses (Culha and Cakırlar 2011).

Seed germination and early seedling growth stages are the beginning of plant life cycle, which are the most vulnerable stages to salinity (Yasar and Yetissin 2023). The salinity causes osmotic stress by lowering the water potential of the germination medium in comparison to the seed interior and reduces or prevents the uptake of water (imbibition process) required for germination and seedling growth (Kaya *et al.* 2012, Moghaddam *et al.* 2018, Kocak *et al.* 2022, Mohammadi *et al.* 2023). In many cases, high salt stress causes absorption of too much Na^+ and Cl^- ions by seeds resulting in ionic stress and ion toxicity to the embryo (Mwando *et al.* 2020) and salinity causes imbalance of hormones especially GA and ABA, and seed nutrients like Na^+ and K^+ (Zaghdoudi *et al.* 2015). High salt concentration in the soil inhibits seed germination or induces dormancy (Farooq *et al.* 2015). Survival of plant under salinity stress is governed by the salt tolerance mechanisms of respective plants to salinity as physiological and biochemical responses (Culha and Cakırlar 2011). Plants can combat salinity stress through enzymatic (such as catalase, ascorbate peroxidase, and superoxide dismutase) and non-enzymatic (such as carotenoids, proline, α -tocopherol, and ascorbic acid) antioxidants (Sharma *et al.* 2012, Singh and Tiwari 2021).

Decrease in fresh and dry weights and biomass of seedling with the increase in the salinity level have been reported

by Nasri *et al.* (2017). General approaches of identifying salinity tolerance sources of crop plants are growing available germplasm in the salt affected soils and selection of desirable plants. However, selection of a salt tolerant genotype based on germination and seedling growth under controlled conditions is simple, quick, precise and very less time consuming (Kaya *et al.* 2012).

Considering these facts, the present study was conducted to evaluate the effects of salinity on the early seedling growth of linseed genotypes under different NaCl levels, with to the aim of identifying potential salt-tolerant genotypes for a varietal improvement program.

Materials and methods

A total of 29 linseed genotypes were tested in the present experiment. Name and sources of the genotypes are shown in Table 1. The genotype, Neela is a linseed variety, which is recommended for cultivation in Bangladesh and it was included in the experiment as a check material. The experiment was conducted in the Laboratory of Oilseed Research Center (ORC) of Bangladesh Agricultural Research Institute (BRAI), Gazipur, Bangladesh during *Rabi* (winter) season of 2019-2020. For germination test, the guidelines recommended by the International Seed Testing Association (ISTA 2003) were followed with minor adjustment according to available of laboratory facilities. Germination and early seedling growth of the genotypes were studied using distilled water (control)

Table 1. Name and source of linseed genotypes studied in the experiment

Name of genotype	Source
Neela (check variety)	Oilseed Research Center (ORC), Bangladesh Agricultural Research Institute (BARI), Gazipur, Bangladesh
BD-7141, BD-7142, BD-7143, BD-7146, BD-10703, BD-10704, BD-10706, BD-10707, BD-10708, BD-10709, BD-10710, BD-10698 and BD-10696	Plant Genetic Resource Center (PGRC), BARI, Gazipur, Bangladesh
Lin-303, Lin-503, Lin-703, Lin-803, Lin-1203, Lin-1303, Lin-1403, Lin-1903, Lin-2403 Lin-1503/2 and Lin-1507/2	Oilseed Research Center, BARI, Gazipur, Bangladesh
Chilmari	Farmer of Chilmari upazila under Kurigram district
Lin-S-19	Farmer of Fatkhira upazila under Shatkhira district
Lin-C-16	China
Lin-V-18	Vietnam
Lin-W-17	Thailand

and under NaCl concentrations of 4, 8, 12, and 16 dS m⁻¹. To prepare salt solutions, a stock solution of NaCl was prepared by dissolving 58.44 gram (1 gram mole) of NaCl in one liter of distilled water. The concentration was measured in an Electrical Conductivity (EC) meter (HANNA) and adjusted to required concentrations of 4, 8, 12 and 16 dS m⁻¹ (1 dS m⁻¹=10 mM) by adding required quantity of distilled water using a pipette under constant stirring.

Before sowing, seeds were surface sterilized with 0.5% sodium hypochlorite solution for 5 minutes followed by rinsing thrice with sterilized distilled water. Surface sterilized seeds were placed on 90 mm diameter filter paper discs in 90 mm Petri dishes. Each Petri dish received 25 seeds and 4 dishes were used for each treatment. The filter paper in a Petri dish was soaked with 10 ml NaCl solutions with desired concentrations and sealed with parafilm to prevent evaporation. Petri dishes containing seeds were incubated in the laboratory at 25-28°C.

The experiment was checked every day at the same time and data were taken on the following parameters:

Germination percentage: Seeds with a radicle of 2 mm were considered as germinated. Five days after seeds were plated into the Petri dishes, germinated seeds were counted and the germination percentage was calculated based on total number of seeds plated.

Five days after germination, 10 seedlings were removed from each Petri dish and data were recorded on the following early seedling growth parameters:

Shoot length and Root length: To determine shoot and root length the distances from crown to leaf tip and root tip were measured as shoot length and root length, respectively.

Seedling fresh weight - After measurement of shoots and root length, the seedlings were soaked with tissue paper and fresh weight of whole seedling was recorded.

Seedling dry weight: After recording the fresh weight, seedlings were dried in an oven at 70°C for constant weight and dry weight was recorded.

Root length to shoot length ratio: The root to shoot ratio was computed by dividing the root length by the shoot length.

Seedling vigor index: Seedling vigor index was calculated using the following equation (Zahedi *et al.* 2011)-

$$SVI = (\text{Germination \%}) \times \text{Mean seedling dry weight in gram}.$$

Salinity tolerance index: The parameter was calculated using the formula suggested by Kokten *et al.* (20210) as:

$$SVI = \frac{\text{Total seedling dry weight at St}}{\text{Total dry weight of seedling at Sc}} \times 100$$

where St is the total seedling dry weight at a salinity level and Sc is the total dry weight of seedling under control.

A 29×5 factorial experiment in completely randomized design with 4 replications was followed, where the first factor was linseed genotypes with 29 levels and the second factor was salinity with 5 levels. The collected data were subjected to statistical analysis for analysis of variance and the treatment means were compared using least significant difference (LSD) test at 5% probability level. For statistical analyses Statistix-10 computer software was used.

Results and discussion

Fresh weight of seedling: Among the 29 genotypes, the fresh weight (FW) was ranged (0.522-0.895 g), (0.476-0.788 g), (0.343-0.673 g), (0.251-0.575) and (0.177-0.472 g) with a mean of 0.710, 0.627, 0.530, 0.438 and 0.341 g at 0, 4, 8, 12 and 16 dS m⁻¹, respectively. The percentage of reduction over control in FW of whole seedlings was 11.69, 25.35, 38.31 and 51.97% at 4, 8, 12 and 16 dS m⁻¹, respectively. The relationship between percent reduction in FW and salinity level was positive, linear and highly significant ($r=0.999^{**}$). In case of genotypes, the heaviest seedling (highest pooled mean of fresh weight) was found in Lin-803 (0.6879 g), which was followed by Lin-W-17 (0.676 g), Lin-2403 (0.672 g), Neela (0.662 g) and Lin-1303 (0.585 g). The lowest fresh weight was recorded from Lin-S-19 (0.353 g) followed by Lin-V-18 (0.385 g), BD-10704 and Lin-1507/2 (Table 2). Results of the present investigation

clearly revealed that salinity stress reduced fresh weight of linseed seedlings. Higher reduction was corroborated with higher levels of salinity. These observations are in

accordance with the findings of Patil *et al.* (2015), Pervez *et al.* (2021), and Kocak *et al.* 2022 for linseed, and Nasri *et al.* 2017 for lettuce, where increase of salinity levels induced seedling fresh weight.

Table 2. Effect of salinity (NaCl) stress on early seedling growth parameters of 29 linseed genotypes under control conditions

Genotype	Seedling fresh weight (g) under different salinity levels (dS m ⁻¹)						Seedling dry weight (g) under different salinity levels (dS m ⁻¹)					
	0	4	8	12	16	Mean	0	4	8	12	16	Mean
Neela	0.864	0.788	0.646	0.552	0.462	0.662	0.294	0.195	0.167	0.108	0.168	0.294
BD-7143	0.574	0.537	0.482	0.376	0.288	0.451	0.129	0.108	0.085	0.054	0.079	0.129
BD-10710	0.722	0.650	0.542	0.473	0.341	0.545	0.182	0.121	0.112	0.104	0.116	0.182
BD-7146	0.758	0.653	0.550	0.480	0.380	0.564	0.211	0.136	0.118	0.106	0.127	0.211
BD-10708	0.780	0.686	0.577	0.486	0.380	0.581	0.170	0.121	0.103	0.085	0.105	0.170
Lin-1303	0.793	0.682	0.577	0.488	0.386	0.585	0.228	0.183	0.115	0.102	0.137	0.228
Lin-703	0.718	0.672	0.547	0.443	0.373	0.550	0.247	0.132	0.116	0.108	0.133	0.247
Lin-303	0.748	0.671	0.552	0.447	0.372	0.558	0.257	0.139	0.117	0.105	0.135	0.257
Lin-1403	0.791	0.640	0.523	0.461	0.379	0.559	0.260	0.177	0.116	0.105	0.144	0.260
Lin-C-17	0.678	0.554	0.462	0.405	0.294	0.479	0.226	0.151	0.115	0.102	0.129	0.226
Lin-1203	0.695	0.644	0.517	0.438	0.323	0.523	0.253	0.145	0.113	0.103	0.135	0.253
Lin-W-17	0.891	0.779	0.668	0.573	0.470	0.676	0.276	0.188	0.146	0.106	0.156	0.276
BD-10698	0.644	0.610	0.494	0.396	0.296	0.488	0.228	0.134	0.111	0.101	0.126	0.228
Lin-503	0.695	0.638	0.549	0.446	0.335	0.533	0.273	0.217	0.116	0.106	0.156	0.273
Lin-C-16	0.736	0.679	0.576	0.452	0.355	0.559	0.250	0.156	0.112	0.106	0.136	0.250
BD-10707	0.705	0.626	0.559	0.431	0.334	0.531	0.224	0.130	0.116	0.102	0.123	0.224
BD-10706	0.668	0.590	0.475	0.361	0.263	0.471	0.209	0.120	0.112	0.101	0.115	0.209
BD-10704	0.582	0.507	0.384	0.291	0.219	0.396	0.215	0.122	0.115	0.106	0.121	0.215
BD-10703	0.694	0.602	0.521	0.425	0.330	0.514	0.232	0.123	0.115	0.105	0.125	0.232
Lin-1507/2	0.564	0.495	0.428	0.310	0.255	0.410	0.222	0.124	0.113	0.104	0.121	0.222
Lin-V-18	0.558	0.486	0.351	0.304	0.228	0.385	0.242	0.121	0.116	0.104	0.126	0.242
Lin-S-19	0.522	0.476	0.343	0.251	0.177	0.353	0.148	0.113	0.100	0.075	0.092	0.148
Lin-2403	0.884	0.775	0.670	0.567	0.465	0.672	0.263	0.173	0.123	0.102	0.142	0.263
Lin-803	0.895	0.781	0.673	0.575	0.472	0.679	0.280	0.186	0.111	0.102	0.149	0.280
Lin-1503/2	0.786	0.632	0.572	0.466	0.368	0.564	0.185	0.140	0.110	0.101	0.114	0.185
Chimari	0.755	0.625	0.558	0.500	0.373	0.562	0.155	0.120	0.110	0.087	0.100	0.155
BD-7141	0.693	0.604	0.540	0.448	0.360	0.529	0.164	0.125	0.106	0.082	0.101	0.164
BD-7142	0.561	0.506	0.483	0.391	0.286	0.445	0.146	0.117	0.102	0.074	0.092	0.146
Lin-1903	0.658	0.608	0.561	0.485	0.337	0.529	0.173	0.126	0.111	0.101	0.112	0.173
Mean	0.710	0.627	0.530	0.438	0.341	0.529	0.218	0.142	0.114	0.098	0.125	0.218
SD	0.103	0.088	0.084	0.082	0.074	0.084	0.045	0.029	0.014	0.013	0.021	0.045
Minimum	0.522	0.476	0.343	0.251	0.177	0.353	0.129	0.108	0.085	0.054	0.079	0.129
Maximum	0.895	0.788	0.673	0.575	0.472	0.679	0.294	0.217	0.167	0.108	0.168	0.294
LSD (P=0.01): Genotype=0.0206, Salinity =0.0008565, Salinity*Salinity = 0.0461							LSD (P=0.01): Genotype=0.0005811, Salinity =0.002413, Salinity*Salinity =0.0130					

Dry weight of seedling: The highest dry weight (DW) of linseed seedlings was recorded under control in all 29 genotypes. Due to salinity stress, the trait was decreased significantly over control. Among the genotypes, the average DW ranged (0.129-0.294 g), (0.108-0.217 g), (0.085-0.167 g) and (0.054-0.108 g) (0.079-0.168 g) with a mean of 0.218, 0.142, 0.114 and 0.098 g seedling⁻¹ under 0, 4, 8 and 12 dS m⁻¹, respectively. As the concentration of NaCl increased, the dry weight of whole seedlings was reduced significantly over control up to 12 dS m⁻¹ showing 36.36, 50.00 and 54.55% reduction at 4, 8 and 12 dS m⁻¹, respectively. In every genotype, the trait was found to be increased significantly at 16 dS m⁻¹ compared 12 dS m⁻¹ showing the average dry weight of 0.125 g seedling⁻¹.

Significantly the highest average DW was found in Neela (0.168 g) followed by Lin-W-17 (0.156 g), Lin-503 (0.156 g), Lin-803 (0.149 g) and Lin-1303 (0.144 g); while the lowest in BD-7143 (0.079 g) followed by BD-7142 (0.092 g), Lin-S-19 (0.092 g), BD-7142 and BD-7143 (Table 2). In the present investigation, it was found that salinity stress reduced dry weight of linseed seedlings and the trait varied with genotypes. Higher reduction was corroborated with higher levels of salinity up to the salinity level of 12 dS m⁻¹. These observations are in accordance with the findings of Kaya *et al.* (2012), Patil *et al.* (2015), Pervez *et al.* (2021) and Yasar and Yetissin (2023) in linseed, and Nasri *et al.* (2017) in lettuce.

Table 3. Effect of salinity (NaCl) stress on seedling root and shoot lengths (cm) of 29 linseed genotypes under control condition

Genotype	Seedling root length (cm) under different levels of salinity (dS m ⁻¹)						Seedling shoot length (cm) under different levels of salinity (dS m ⁻¹)					
	0	4	8	12	16	Mean	0	4	8	12	16	Mean
Neela	4.25	4.90	3.08	2.11	1.30	3.12	7.20	6.70	4.40	3.33	1.53	4.63
BD-7143	3.88	3.43	4.48	3.95	3.18	3.78	5.73	4.90	5.50	4.83	3.68	4.93
BD-10710	4.40	3.78	3.38	3.10	2.40	3.41	5.70	5.38	4.48	3.73	3.28	4.51
BD-7146	5.13	4.08	3.83	3.30	2.88	3.84	5.68	5.18	4.88	4.10	3.48	4.66
BD-10708	5.18	3.90	3.75	3.00	2.70	3.66	5.45	4.80	4.08	3.68	3.13	4.23
Lin-1303	4.93	4.18	3.58	2.45	2.18	3.51	6.43	5.05	3.38	1.45	1.03	3.47
Lin-703	3.68	3.88	3.40	2.43	1.93	3.06	5.78	4.50	4.23	3.03	1.20	3.75
Lin-303	3.75	3.33	3.35	2.20	1.78	2.88	4.70	4.53	4.00	1.50	0.93	3.13
Lin-1403	4.48	3.78	3.28	2.43	1.78	3.15	5.33	5.38	2.50	1.23	0.98	3.08
Lin-C-17	4.13	2.93	2.24	1.83	1.84	2.59	4.93	3.53	2.85	1.73	0.96	2.80
Lin-1203	2.80	2.29	2.25	1.80	1.25	2.08	4.48	3.11	1.35	0.91	0.76	2.12
Lin-W-17	4.26	2.84	2.51	2.81	1.30	2.74	6.08	4.48	3.13	1.75	0.99	3.28
BD-10698	2.05	1.49	1.21	1.00	1.38	1.43	3.84	2.62	1.71	1.15	0.89	2.04
Lin-503	2.42	2.28	2.36	1.69	1.20	1.99	3.81	2.83	1.45	1.04	0.66	1.96
Lin-C-16	3.33	2.19	1.98	1.56	1.13	2.04	3.54	2.48	1.21	1.06	0.60	1.78
BD-10707	2.52	2.12	1.25	0.98	1.10	1.59	4.35	3.86	1.96	0.94	0.67	2.35
BD-10706	2.88	2.06	1.67	1.13	0.82	1.71	4.13	3.11	1.87	0.98	0.72	2.16
BD-10704	2.59	1.95	1.62	1.02	0.74	1.58	4.28	2.93	1.43	0.93	0.67	2.05
BD-10703	2.31	2.52	2.33	1.80	1.26	2.04	3.19	2.85	1.93	0.95	0.63	1.91
Lin-1507/2	2.86	2.50	1.95	1.40	1.10	1.96	3.75	2.96	1.73	0.92	0.58	1.99
Lin-V-18	3.20	2.36	1.90	1.43	0.98	1.97	3.90	3.30	1.88	1.05	0.73	2.17
Lin-S-19	3.00	2.30	1.90	1.40	1.01	1.92	3.48	2.55	1.70	1.05	0.78	1.91
Lin-2403	3.83	2.93	2.42	1.54	2.14	2.57	6.45	4.40	2.55	1.42	1.00	3.16
Lin-803	2.65	2.18	2.21	1.93	2.10	2.21	5.25	3.95	2.78	2.11	2.12	3.24
Lin-1503/2	3.57	2.99	2.64	2.56	2.34	2.82	4.68	4.28	2.99	2.93	2.06	3.39
Chimari	3.67	3.87	4.03	3.21	2.60	3.47	5.34	4.58	3.83	3.65	1.70	3.82
BD-7141	4.06	3.52	3.53	2.88	1.83	3.16	5.83	5.33	4.93	4.28	2.90	4.65
BD-7142	4.10	4.24	3.05	3.05	2.55	3.40	5.80	5.38	3.83	3.63	2.97	4.32
Lin-1903	4.43	3.68	3.43	3.03	2.65	3.44	5.83	5.25	4.43	3.65	2.15	4.26
Mean	3.60	3.05	2.71	2.17	1.77	2.66	5.00	4.14	3.00	2.17	1.51	3.16
SD	0.87	0.87	0.87	0.81	0.69	0.75	1.04	1.11	1.28	1.30	1.01	1.04
Minimum	2.05	1.49	1.21	0.98	0.74	1.43	3.19	2.48	1.21	0.91	.058	1.78
Maximum	5.18	4.90	4.48	3.95	3.18	3.84	7.20	6.70	5.50	4.83	3.68	4.93
LSD (P=0.01):Genotype=0.2603, Salinity =0.1081, Salinity *Salinity = 0.5822							LSD (P=0.01):Genotype=0.2902, Salinity =0.1205, Salinity *Salinity = 0.6489					

Root length of seedling: The root length (RL) of seedling was found to be increased in 4 genotypes (Neela, Lin-703, BD-10703 and Chilmari) at 4 dS m⁻¹ compared to control but it was begun to decrease from immediate higher level (8 dS m⁻¹). In other 25 genotypes, the RL was begun to decrease from the lowest salinity level (4 dS m⁻¹) and continued up to the highest level (16 dS m⁻¹). In these 25 genotypes the longest root was recorded under control and use of NaCl solution for soaking germination paper caused gradual reduction in the trait over control in every genotype. The average RL among these 25 genotypes ranged (2.05-4.24 cm), (1.49-4.24 cm), (1.21-4.48 cm), (0.98-3.95 cm) and (0.74-3.18 cm) with a mean of 3.61, 2.93, 2.563, 2.14 and 1.77 cm at 0, 4, 8, 12 and 16 dS m⁻¹, respectively. The reduction of the trait was significant at all salinity levels in 16 genotypes and at 8, 12 and 16 dS m⁻¹ in the rest 13 genotypes. The reduction in pooled mean of RL over control was 15.30, 24.76, 39.69 and 50.77% at 4, 8, 12 and 16 dS m⁻¹, respectively. The relationship between percentage of reduction and level of salinity was linear, positive and highly significant ($r=0.996^{**}$). In case of genotypes, the longest root was found in BD-10708 (5.18 cm) followed by BD-7146 (5.13 cm), Lin-1303 (4.93 cm), Lin-1403 (4.48 cm), Lin-1903 (4.43 cm) and BD-10710 (4.40 cm). The shortest root was found in BD-10698 (2.05 cm) followed by BD-10703 (2.31 cm), Lin-503 (2.42 cm) and BD-10707 (2.52 cm) under control (Table 3). Roots are the first organs to be exposed to salinity and this organ of plant is in direct contact with the soil. They absorb water and nutrient from the soil and transfer it to shoots. Salinity prevents the maintenance of nutrient levels necessary for plant growth and limits root development and seedling growth (Abari *et al.* 2011, Bilgili *et al.* 2011). It was also reported that increasing salt doses decreased root lengths (Moghaddam *et al.* 2018, Kocak *et al.* 2022).

Shoot length of seedling: In all genotypes, the maximum shoot length (SL) was found under control and salinity stress caused reduction in the trait in parallel with increasing salinity levels. At the salinity levels of 0, 4, 8, 12 and 16 dS m⁻¹, the SL was varied (3.19-7.20 cm), (2.48-6.70 cm), (1.21-5.50 cm), (0.91-4.83 cm) and (0.58-3.68 cm) with a mean of 5.00, 4.14, 3.00, 2.17 and 1.51 cm, at 4, 8, 12 and 16 dS m⁻¹, respectively. The reduction in the trait was begun at 4 dS m⁻¹ in 17 genotypes, and at 8 dS

m⁻¹ in other 12 genotypes, and in both cases the reduction was continued up to 16 dS m⁻¹. In every genotype, the pooled mean of shoot length was corroborated with the level of salinity showing the shortest shoot at the highest level. Percentage of reduction in shoot length over control was 17.20, 40.00, 56.60 and 69.80% at 4, 8, 12 and 15 dS m⁻¹, respectively. The relationship ($r=0.992^{**}$) between reduction percentage in the trait and level of salinity was linear, positive and significant. Withstanding the salinity stress, genotype BD-7143 showed the longest shoot (4.93 cm) followed by BD-7146 (4.66 cm), BD-7141 (4.65 cm), Neela (4.63 cm) and BD-10710 (4.51 cm); while the lowest was recorded from Lin-C-16 (1.78 cm), which was statistically similar to BD-10703 (1.91 cm), Lin-S-19 (1.91 cm), Lin-503 (1.96 cm), Lin-1507/2, BD-10698 and BD-10704 (Table 3). Results of the present investigation showed that salinity stress reduced shoot lengths of linseed seedlings. Higher reduction was corroborated with higher levels of salinity. When a plant is grown under saline environment, reduction in shoot length is common in various crop species (Amin *et al.*, 1996). The findings supported the findings of many other researchers (Nasri *et al.* 2017, Pervez *et al.* 2021, Kocak *et al.* 2022, Yasar and Yetissin 2023). They also found significant reduction in fresh and dry weights of seedlings and whole seedling biomass due to salinity stress.

Root to shoot ratio of seedling: Seven genotypes (Neela, BD-7143, BD-10710, BD-7146, BD-10708, BD-7141 and BD-714) showed statistically similar root to shoot ratio (RL:SL) values up to the highest level, while in other 7 genotype (Lin-1403, Lin-1203, Lin-503, Lin-C-16, BD-10703, Lin-1507/2, Lin-S-19), the trait was significantly increased at all salinity levels compared to control. The RL:SL was statistically similar up to 12 dS m⁻¹ and significantly increased over control at 16 dS m⁻¹ in other 6 genotypes (Lin-703, BD-10698, Lin-803, Lin-1503/2, Chilmari, Lin-1903). In 9 genotypes (Lin-1303, Lin-303, Lin-C-17, Lin-W-17, BD-10707, BD-10706, BD-10704, Lin-V-18, Lin-2403), the RL: SL was statistically similar up to 8 dS m⁻¹ but increased significantly over control at 12 and 16 dS m⁻¹ indicating increase in root length under salinity stress at two higher levels in these 9 genotypes. Higher RL:SL indicated higher growth of root length compared to shoot length under salinity stress condition. The pooled mean of RL: SL under control (0.73) and at 4

dS m⁻¹ (0.75) was almost identical but significantly lower compared to 3 higher levels of salinity. Significantly the highest RL:SL was found at 16 dS m⁻¹ (1.47) followed by 12 dS m⁻¹ (1.20) and 8 dS m⁻¹ (1.00). Due to salinity stress, the trait was increased by 2.74, 36.99, 64.38 and 101.37% at 4, 8, 12 and 16 dS m⁻¹. Regarding response of RL:SL of 29 genotypes, the pooled mean in 14 genotypes (BD-10703, Lin-C-16, Lin-503, Lin-1403, Lin-1303, Lin-1203, Lin-1507/2, Lin-303, Lin-S-19, Lin-C-17, Lin-2403, Lin-V-18, Lin-W-17, Chimera) was more than 1.00 (1.02-1.55) and the rest 15 genotypes showed less than 1.00 (0.70-0.97) RL:SL values. Well defined variations were observed in linseed genotypes in terms of root to shoot ratio (Table 4).

In general, both root and shoot lengths of seedlings were found to be diminished due to salinity stress in all genotypes in the present study. However, the length of root was higher compared to shoot in 14 genotypes under higher salinity levels where the RL:SL was more than 1.00. Similar findings have been reported by Pervez *et al.* (2021), Kocak *et al.* (2022) and Yasar and Yetissin (2023). Keaya *et al.* (2012) tested 10 linseed genotypes for their response to salinity stress and found that most genotypes showed increase in root length. According to Pervez *et al.* (2021) common response of linseed genotypes are to enhance root to shoot to ratio depending of variations in genotypes.

Seedling vigor index of seedling: In every genotype, the highest seedling vigor index (SV) was found under control (0 dS m⁻¹), which was reduced over control due to salinity stress. Higher reduction was corroborated with higher dose of NaCl. The SVI values among the genotypes ranged (1.78 - 28.86), (8.37 - 19.58), (6.06 - 14.16), (3.13 - 8.96) and (0.92-5.11) with a mean of 19.92, 11.98, 8.91, 6.58 and 2.54 at 0, 4, 8, 12 and 16 dS m⁻¹, respectively. The reduction in SVI was increased significantly in parallel with increasing salinity levels. The average SVI in 29 genotypes was reduced by 37.27, 52.21, 64.28 and 84.57% respectively at 4, 8, 12 and 16 dS m⁻¹. Withstanding the salinity stress, the highest average mean of SVI was recorded from Neela followed by Lin-W-17, Lin-503, Lin-1403 and Lin-2403, while the lowest was found in BD-7143 (6.06) followed by Chilmari (6.46)

and BD-7142. Genotypes Neela, Lin-W-17, Lin-503, Lin-1403 and Lin-2403 were noted as better performers showing vigorous seedlings (Table 4). These findings of the present investigations are in agreement with the reports of many other researchers (Patil *et al.* 2015, Pervez *et al.* 2021, Yasar and Yetissin 2023), who reported that the SVI and RL:SL ratio of linseed decreased as salinity stress increased. They also reported that vigor index is a useful trait to identify salt resistant genotypes.

Salt tolerant index of seedling dry weight: In all genotypes, the salt tolerance index (STI) was significantly decreased over control starting from the lowest level (4 dS m⁻¹) and continued up to the highest level (16 dS m⁻¹). The reduction was gradual with increasing NaCl concentrations. The mean STI was reduced by 33.82, 46.08, 54.14 and 77.34% over control at 4, 8, 12 and 16 dS m⁻¹, respectively. The results of ANOVA indicated existence of variations among the genotypes in terms of SVI, which were influenced by salinity stress. Among 29 genotypes, the average STI was varied from 52.28 to 54.84% with a grand mean of 57.72%. Ten genotypes showed more than 60% STI and the best performing genotype was Lin-1903 followed by Chilmari, BD-10710, BD-7142 and Lin-S-19 (Table 5). The results of the present investigation showed that the STI was reduced significantly in parallel with increasing salinity levels. The findings are in agreement with the reports of Pervez *et al.* (2021) and Yasar and Yetissin (2023) who also identified adverse effect of salinity on salt tolerance indices of linseed.

In the present investigation, gradual decrease in fresh and dry weights and lengths of root and shoot of seedlings with increasing salinity levels was observed. Wide ranges of variations were found in the responses of early seedling growth related traits of linseed genotypes to salinity stress. Similar findings have also been reported by Pervez *et al.* (2021) and Yasar and Yetissin (2023). Variation in varietal responses to salinity stress in linseed has been reported by Pervez *et al.* (2021), Kocak *et al.* (2022) and Yasar and Yetissin (2023). Salt tolerant and survival of plant genotypes under salt stress might be attributed to their various salt resistance mechanisms such as prevention of the entry of salts into the plant, tissue tolerance, compartmentalization of toxic salt in

Table 4. Effect of salinity (NaCl) stress on root length:shoot length ratio and seedling vigor index of 29 linseed genotypes under control conditions

Genotype	Root length to shoot length ratio (RL:SL) under different salinity levels (dS m ⁻¹)						Seedling vigor index (SVI) under different salinity levels (dS m ⁻¹)					
	0	4	8	12	16	Mean	0	4 dS	8 dS	12	16	Mean
Neela	0.59	0.73	0.71	0.64	0.84	0.70	28.86	16.86	14.16	8.96	5.11	14.79
BD-7143	0.67	0.73	0.83	0.82	0.86	0.78	11.78	8.42	6.06	3.13	0.92	6.06
BD-10710	0.78	0.70	0.76	0.83	0.73	0.76	16.64	10.04	7.44	5.52	1.33	8.19
BD-7146	0.91	0.79	0.79	0.81	0.83	0.83	19.30	11.52	9.40	5.45	2.48	9.63
BD-10708	0.90	0.81	0.92	0.82	0.86	0.87	16.10	10.85	8.72	7.25	3.72	9.33
Lin-1303	0.81	0.83	1.06	1.84	2.19	1.35	21.99	16.12	8.97	5.11	2.73	10.99
Lin-703	0.64	0.86	0.80	0.82	1.62	0.95	22.60	10.74	8.66	5.56	3.00	10.11
Lin-303	0.80	0.74	0.84	1.50	1.97	1.17	23.51	11.79	9.13	5.61	2.78	10.56
Lin-1403	0.85	0.71	1.36	2.02	1.81	1.35	25.16	16.25	9.88	8.71	5.06	13.01
Lin-C-17	0.85	0.84	0.78	1.15	1.97	1.12	17.37	10.80	7.44	5.46	2.41	8.70
Lin-1203	0.63	0.74	1.67	1.99	1.65	1.33	24.48	13.02	9.58	8.41	3.52	11.80
Lin-W-17	0.71	0.64	0.81	1.68	1.34	1.03	25.71	16.56	11.88	8.61	4.10	13.37
BD-10698	0.53	0.58	0.73	0.92	1.50	0.85	22.42	12.24	9.78	8.44	3.43	11.26
Lin-503	0.68	0.81	1.67	1.63	1.98	1.36	25.84	19.58	9.65	8.46	3.14	13.33
Lin-C-16	0.95	0.88	1.75	1.55	2.05	1.44	20.38	12.03	8.23	5.48	2.41	9.71
BD-10707	0.58	0.55	0.64	1.07	1.46	0.86	21.30	11.49	9.84	8.29	2.02	10.59
BD-10706	0.71	0.67	0.92	1.16	2.78	0.95	18.50	10.00	8.80	7.22	2.09	9.32
BD-10704	0.60	0.68	1.16	1.18	1.26	0.97	19.66	10.35	9.40	7.98	2.20	9.92
BD-10703	0.77	0.91	1.41	1.89	1.21	1.55	19.73	9.60	8.65	6.98	2.86	9.56
Lin-1507/2	0.78	0.85	1.15	1.53	1.93	1.25	20.38	10.75	9.23	7.98	2.43	10.16
Lin-V-18	0.83	0.72	1.07	1.39	1.40	1.08	22.94	10.44	9.31	5.70	1.66	10.01
Lin-S-19	0.87	0.92	1.16	1.34	1.35	1.13	13.28	9.56	8.12	5.96	1.28	7.64
Lin-2403	0.59	0.67	0.97	1.21	2.14	1.12	24.96	15.24	10.41	8.29	2.63	12.30
Lin-803	0.50	0.56	0.80	0.95	1.17	0.80	23.30	13.92	6.82	5.46	2.55	10.41
Lin-1503/2	0.76	0.71	0.87	0.87	1.15	0.88	17.23	11.40	9.16	8.04	1.81	9.53
Chimari	0.70	0.85	1.08	0.92	1.54	1.02	11.85	8.37	6.57	4.18	1.32	6.46
BD-7141	0.70	0.66	0.75	0.71	0.76	0.71	14.72	10.17	7.91	4.92	1.63	7.87
BD-7142	0.71	0.79	0.81	0.88	0.87	0.81	13.57	9.71	7.78	4.53	1.23	7.36
Lin-1903	0.76	0.70	0.77	0.83	1.26	0.86	14.09	9.62	7.37	5.19	1.94	7.64
Mean	0.73	0.75	1.00	1.20	1.47	1.03	19.92	11.98	8.91	6.58	2.54	9.99
SD	0.12	0.10	0.31	0.42	0.52	0.22	4.61	2.83	1.60	1.64	1.06	4.42
Minimum	0.50	0.55	0.64	0.64	0.73	0.70	11.78	8.37	6.06	3.13	0.92	6.06
Maximum	0.95	0.92	1.75	2.02	2.78	1.44	28.86	19.58	14.16	8.96	5.11	14.79
LSD(P≤0.05): Genotype=0.1808, Salinity =0.0751, Salinity*Salinity =0.4043							LSD(P≤0.05): Genotype=0.6592, Salinity =0.2737, Salinity*Salinity =1.4741					

the vacuoles without any physiological impairment, ion discrimination (K^+ , Na^+ , Cl^- and SO_4^{2-}) during uptake by roots and translocation to shoot. Various biochemical processes such as production of some enzymes, hormones, antioxidants, osmoprotectants (malate, proline) are also involved in salt mechanisms of crop plants (Almansouri *et al.* 2001; Parida and Das 2005).

Conclusion

The results of the present investigation revealed that salinity (NaCl) stress at 4, 8, 12 and 16 dS m⁻¹ caused significant reduction over control in early seedling growth related parameters such as fresh and dry weights, and root length and shoot length of seedling, root to shoot ratio,

Table 5. Effect of salinity (NaCl) stress on salt tolerance indices of seedling dry weight of 29 linseed genotypes under control condition

	Salt tolerance index (%) under different levels of salinity (dS m ⁻¹)					Mean
	0	4	8	12	16	
Neela	100.00	66.27	56.73	36.63	26.06	57.14
BD-7143	100.00	83.66	65.76	41.63	15.17	61.24
BD-10710	100.00	66.39	61.43	57.02	33.88	63.74
BD-7146	100.00	64.37	55.82	50.12	30.64	60.19
BD-10708	100.00	71.09	60.47	50.29	27.43	61.86
Lin-1303	100.00	80.22	50.33	44.83	26.59	60.39
Lin-703	100.00	53.35	46.85	43.61	26.16	53.99
Lin-303	100.00	54.09	45.42	40.93	22.42	52.57
Lin-1403	100.00	68.11	44.67	40.15	24.30	55.45
Lin-C-17	100.00	66.86	50.98	45.52	21.66	57.00
Lin-1203	100.00	57.71	45.03	41.06	23.96	53.55
Lin-W-17	100.00	68.06	52.81	38.29	23.41	56.51
BD-10698	100.00	58.84	49.02	44.78	24.22	55.37
Lin-503	100.00	78.03	43.00	39.27	24.58	56.98
Lin-C-16	100.00	60.97	45.24	42.79	22.29	54.26
BD-10707	100.00	58.05	51.69	45.27	18.46	54.69
BD-10706	100.00	57.39	53.65	48.19	15.86	55.02
BD-10704	100.00	56.78	53.61	49.41	21.24	56.21
BD-10703	100.00	52.93	49.81	45.28	21.82	53.97
Lin-1507/2	100.00	55.88	50.90	46.87	19.65	54.66
Lin-V-18	100.00	49.90	48.13	42.86	20.50	52.28
Lin-S-19	100.00	76.27	67.46	50.51	17.96	62.44
Lin-2403	100.00	65.65	46.62	38.63	54.18	61.02
Lin-803	100.00	66.37	39.53	36.49	53.45	59.17
Lin-1503/2	100.00	75.61	59.62	54.47	62.00	70.34
Chimari	100.00	77.35	70.87	55.98	64.79	73.80
BD-7141	100.00	76.15	64.52	50.15	62.02	70.57
BD-7142	100.00	80.07	69.76	50.51	63.16	72.70
Lin-1903	100.00	72.75	64.06	58.26	64.87	71.99
Mean	100.00	66.18	53.92	45.86	57.72	59.62
SD	100.00	83.66	70.87	58.26	64.87	6.48
Minimum	0.00	9.58	8.61	6.05	4.03	52.28
Maximum	100.00	49.90	39.53	36.49	52.28	73.80
LSD (P=0.05): Genotypes = 1.3913, Salinity = 0.5777, Genotypes*Salinity = 0.4043						

seedling vigor index and salt tolerant index. The reduction was gradual in parallel with increased level of salinity. A wide range of variations were observed in the genotypes of linseed in terms of responses of early seedling growth related parameters to salinity stress. The genotypes Neela, Lin-W-17, B-10708, BD-10710, Lin-1303, Lin-2403, BD-10698, Lin-S-19, BD-7143 and Lin-503 demonstrated considerably early seedling growth and salt tolerance, withstanding salinity stress effectively. These genotypes were noted as prospective materials for the development of salt tolerant linseed varieties.

Acknowledgements

The first author gratefully acknowledges the Kshrichi Gobeshana Foundation, Bangladesh Agricultural Research Council, Dhaka, Bangladesh and ACIAR, Australia for providing fund for the research project and for granting financial assistance. The first author would like to express her gratitude to the Director-General, Bangladesh Agricultural Research Institute (BARI), and Director, Oilseed Research Center, BARI, Gazipur, Bangladesh for granting deputation to work in the project. She also gratefully acknowledges the Genetic Resource Division of BARI for providing the linseed genotypes.

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